

MAJOR PROJECT BRIEF

CineShield 2.0

An AI-powered copyright threat intelligence system for movies, web series, TV shows, OTT originals, clips, and other premium digital video.

215B+

GLOBAL PIRACY VISITS

MUSO reports piracy website visits measured in 2024.

INR224B

INDIA PIRACY ECONOMY

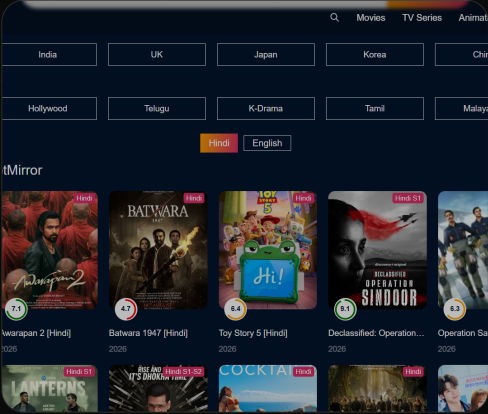
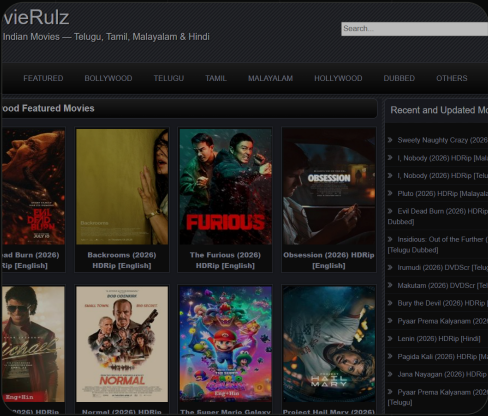
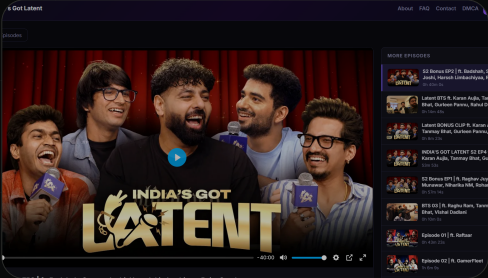
EY-IAMAI estimate for India in 2023.

63%

STREAMING SHARE

EY-IAMAI: streaming was 63% of pirated-content access sources in India.

Problem: content is no longer copied only as a file. It appears as transformed streams, mirror uploads, cropped clips, subtitle edits, and platform-specific re-posts.

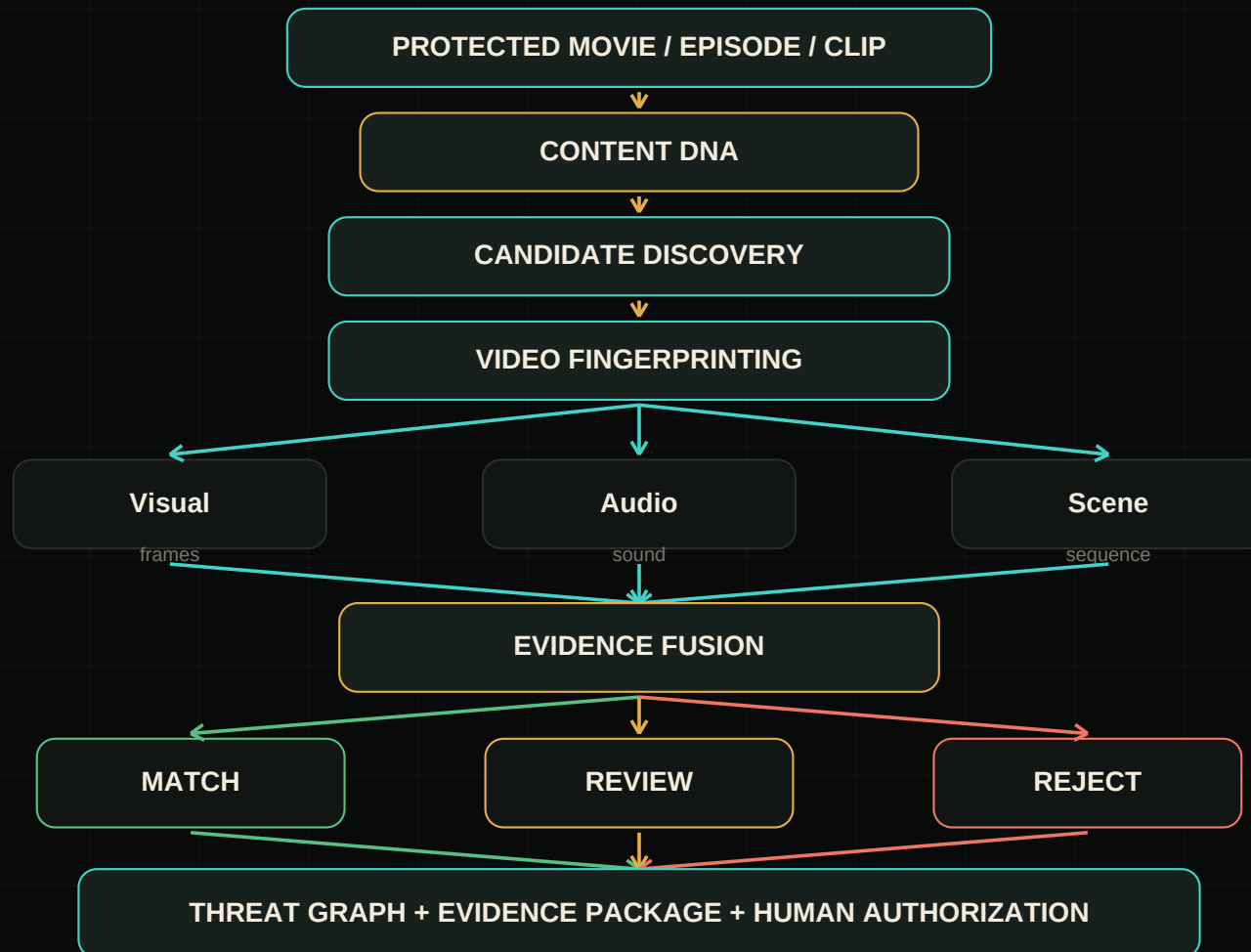


Screenshots are used only as problem context examples.

THE IDEA

Content DNA, not filename search

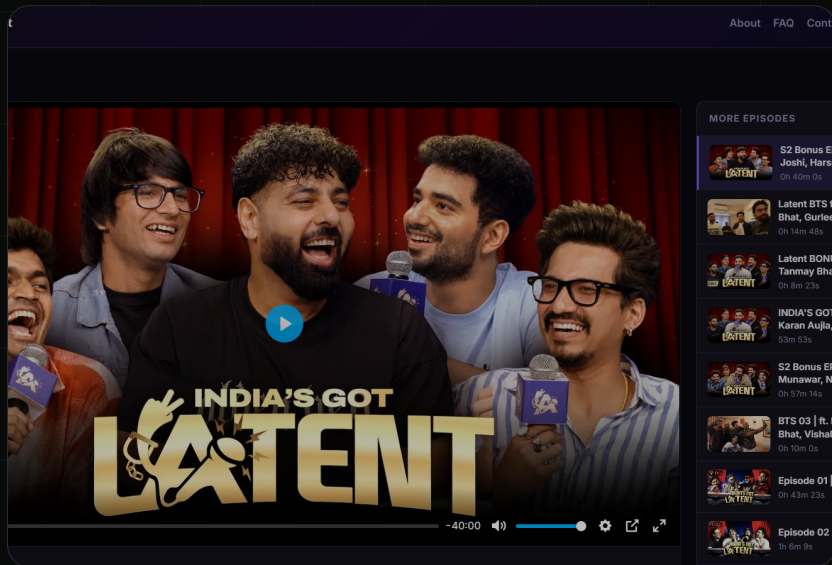
A one-glance architecture for verifying transformed copies and turning results into review-ready evidence.



EXAMPLE SCENARIO

An OTT episode leak becomes an evidence case

The demo story can be explained without heavy technical language.



Example: a released episode or show clip starts appearing on unauthorized viewing pages.

1. Register

Rights holder uploads the original episode, show, trailer, or event video.

2. Generate DNA

The system samples frame, scene, audio/temporal, and metadata signals.

3. Process candidates

Uploaded files, partner feeds, or lawful URL queues become candidates.

4. Verify

Transformed copies are scored using adaptive multimodal fusion.

5. Respond safely

The output is an evidence package for human authorization.

Important boundary: CineShield produces evidence and review priority. It does not declare legality or perform automatic takedowns.

A real pipeline, not only a dashboard

The UI is the command center; the backend processes local media and returns measured scan results.

Implemented now

- Frontend upload flow for protected video and candidate videos
- FastAPI backend with /api/scan and /api/scan/demo
- OpenCV/NumPy fingerprint extraction from local media
- Adaptive comparison: visual, scene, temporal, duration, coverage
- Candidate ranking, confidence, reasons, and investigator summary
- Synthetic benchmark plus real-media smoke test

Tech stack

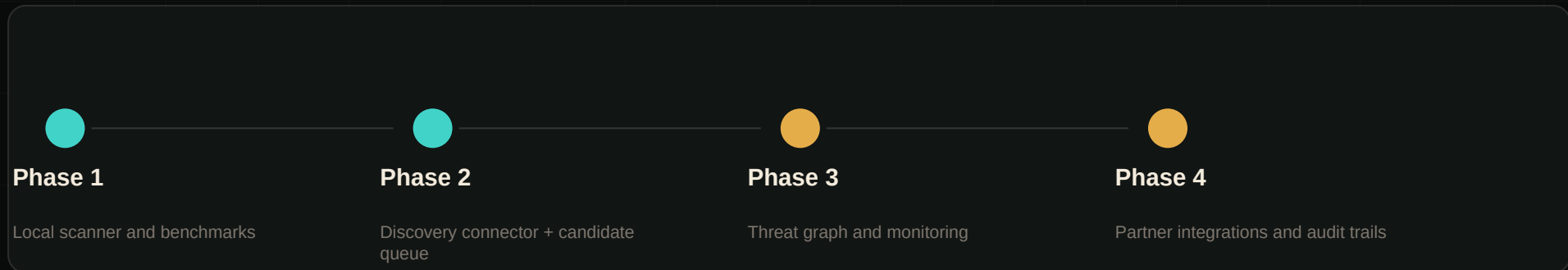
- Frontend: HTML, CSS, JavaScript dashboard
- Backend: Python FastAPI service
- Media engine: OpenCV, NumPy, frame sampling
- AI/ML: CV features plus adaptive evidence fusion
- Investigator: priority and uncertainty from structured evidence
- Safety: human review gate before response action

Demo path: upload original -> upload suspected copies -> scan -> rank -> inspect evidence -> generate review package.

ROADMAP

Discovery connects into the same DNA engine

The next stage is not rebuilding verification. It is adding lawful source connectors and queue-based monitoring.



Not implemented yet

- Internet-wide discovery at scale
- Continuous external monitoring
- Production takedown integrations
- Automated legal decisions

Expected outcome

A practical content-protection system that can help rights holders identify transformed video copies, prioritize review, and preserve an audit trail for safe action.

Sources: MUSO Piracy by Industry, EY-IAMAI The Rob Report press release, FICCI-EY India M&E; report, U.S. Chamber copyright report. The 63% metric refers to streaming's share among pirated-content access sources in India. Screenshots supplied by project owner as problem context.